Pre-survey

1. Scan the QR code
2. Select ‘Teaching Sign Language with AI’ for the workshop you’re attending

We encourage you to take the survey; the data will be valuable for our research and for improving future workshops. 😁

Ivana Hernandez, Keren Zhang, Joey Chen

Meet Us

Emu Unicorn Sauce (EUS)

Joey - Software Dev major

Worked on obtaining the required code for the project.

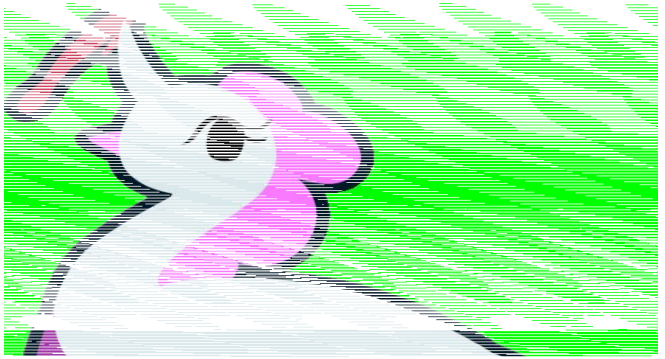
Ivana - Software Dev major

Worked on/trained the AI and the grading system.

Drew the logo for Gesture Control and the EUS mascot

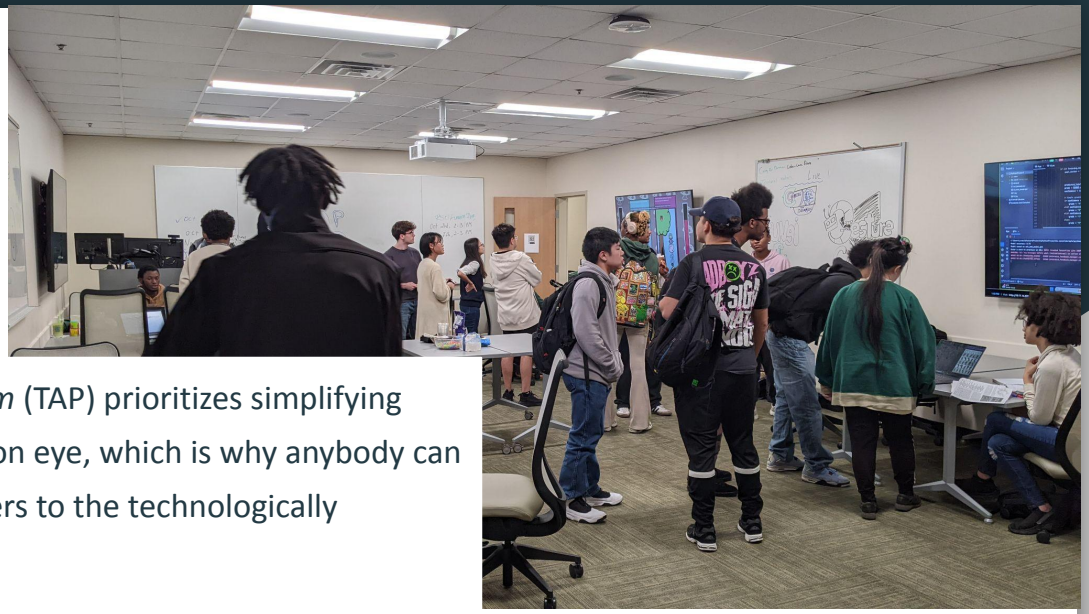
Keren - Data Science major

Worked on integrating the code into a website with both front-end and back-end.



Emu Unicorn Sauce mascot, animated by Ivana

About TAP



- The *Technology Ambassador Program* (TAP) prioritizes simplifying technology in the view of the common eye, which is why anybody can enroll: from experienced programmers to the technologically illiterate.
- Students will choose a project to work on during the semester and maintain motivation consistently to progress through all checkpoints.
- Outside the learning of technical skills needed to work on your project, TAP also leads to developing self-discipline and soft skills, which are arguably just as important!

Image from 10/10/25 TAP EXPO

Our Goals

- Through *Gesture Control,* we can showcase a real-world use of AI through the idea of teaching sign language to the user.
- Users will learn how to sign a given word by the individual letters in the *American Sign Language* (ASL); the program grades you on your accuracy.
- AI, especially outside of LLMs, may seem a little alien when compared to other forms of technology. Hopefully we can demystify AI, as this involves basic machine learning that anybody can learn.

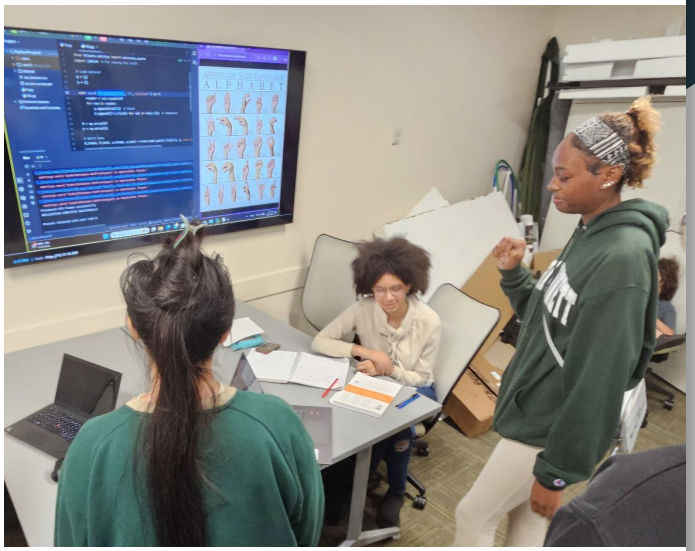
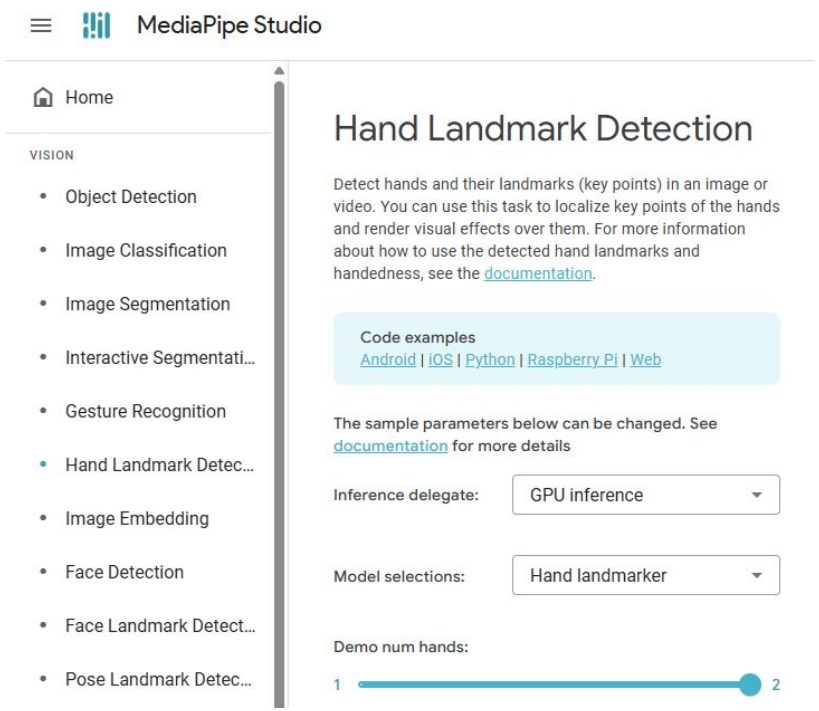


Image from 10/10/25 TAP EXPO

Hand Recognition

- ‘Hand Landmark’ code is from MediaPipe, trained off thousands of hands https://mediapipe-studio.webapps.google.com/studio/demo/hand_landmarker
- It divides the hand into 21 points, with each point having its own coordinates



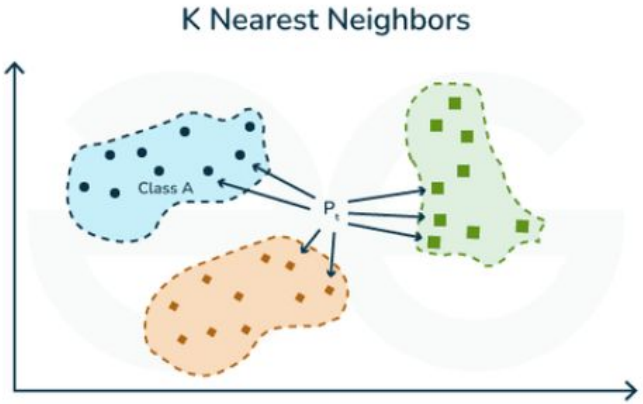
The AI

- Using the MediaPipe code, we had each hand as a set of coordinates to a corresponding letter into a CSV file.
- 15 samples for each letter was collected (though ideally, you would need hundreds for accuracy)
- The CSV file was used to train the AI for ‘letter recognition’

A	0.5863040	0.5377089	-5.82E-07	0.6650956	0.514521	-0.02117
P	0.631982	0.4389705	1.6512025	0.6545878	0.3721058	-0.02808
P	0.6509067	0.4070387	2.3899721	0.6673443	0.3489736	-0.02832
L	0.5822718	0.6613953	1.2963180	0.650313	0.6372552	-0.02226
E	0.6785417	0.5674809	5.4717407	0.7542836	0.5116733	0.023292

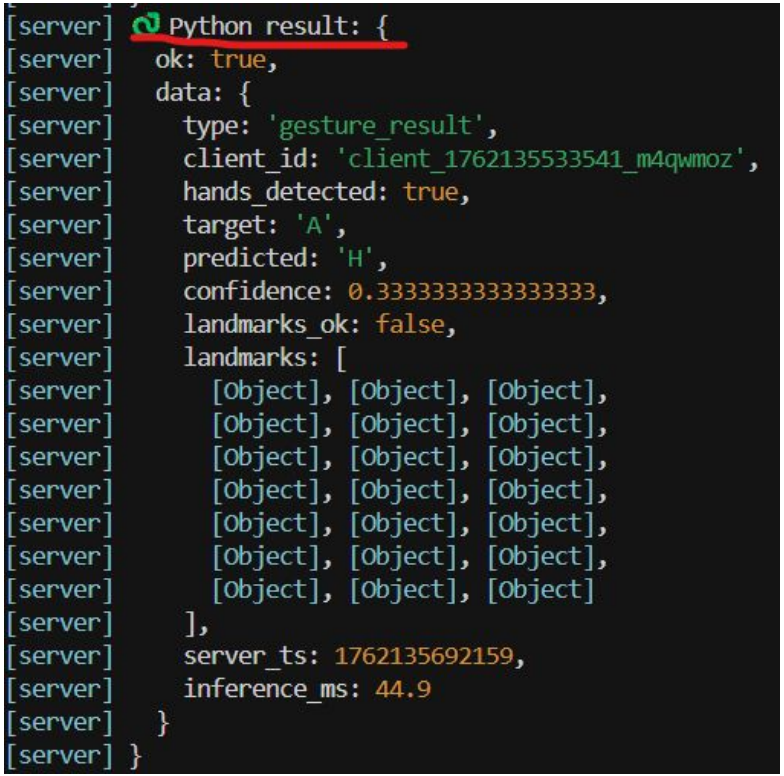
KNN (K-Nearest Neighbors)

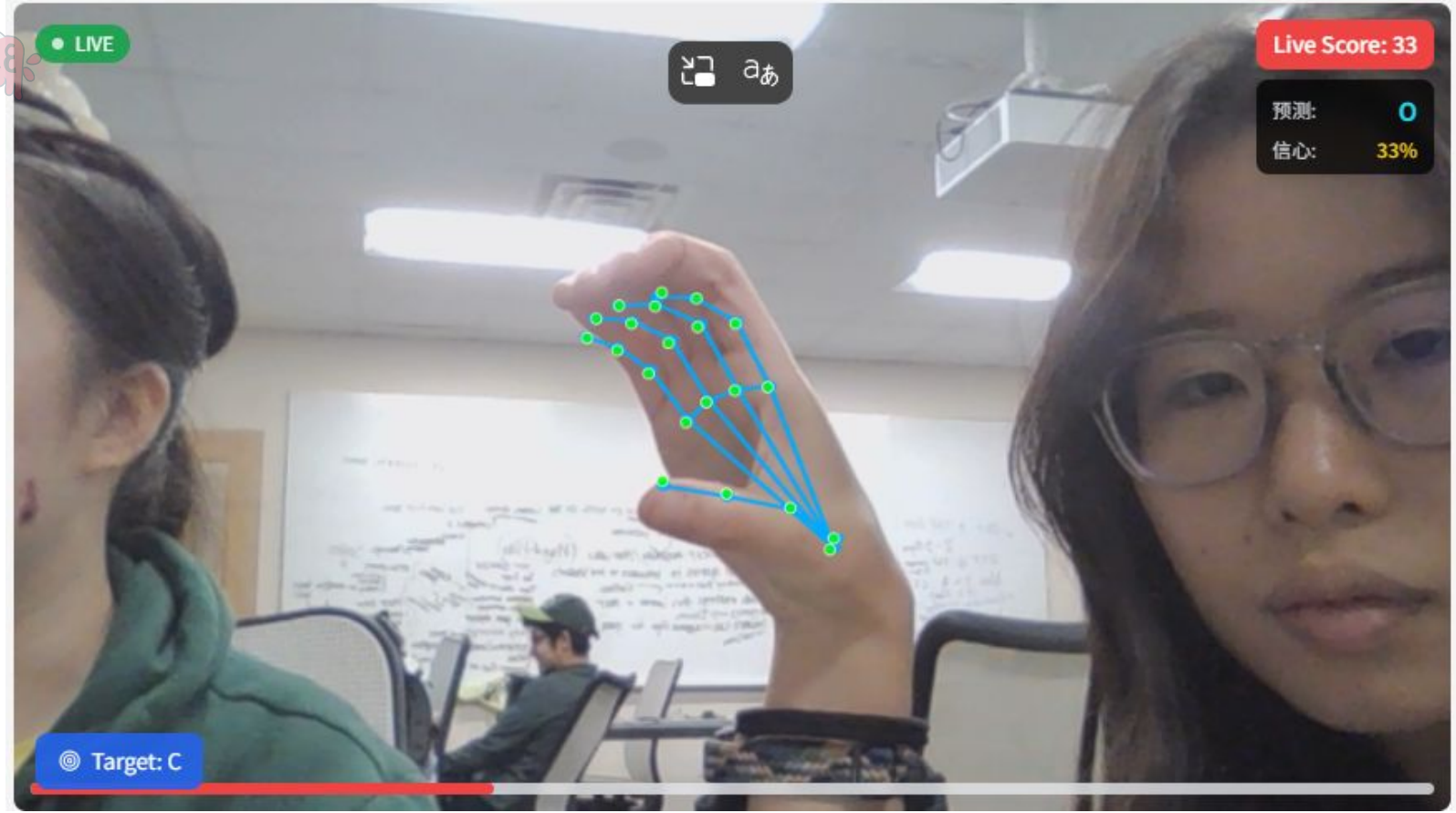
- K-Nearest Neighbors (KNN) is a simple supervised algorithm. In this case, ideal for classification.
- As the user's hands come into frame, the live coordinates for a given letter are tested against the coordinates in the dataset.
- Finally, a basic 'if-then' A-D grading system based off how close you were to the predicted letter.



Setting the AI model to run as our backend

- AI doesn't "see" a picture like humans. It tracks 21 hand **joint points** → analyzes **geometry** → makes a **guess**.
- Low confidence or bad points = we **ignore** that frame
- AI is smart, but it also doubts itself. We need to **check confidence** before trusting its answer.



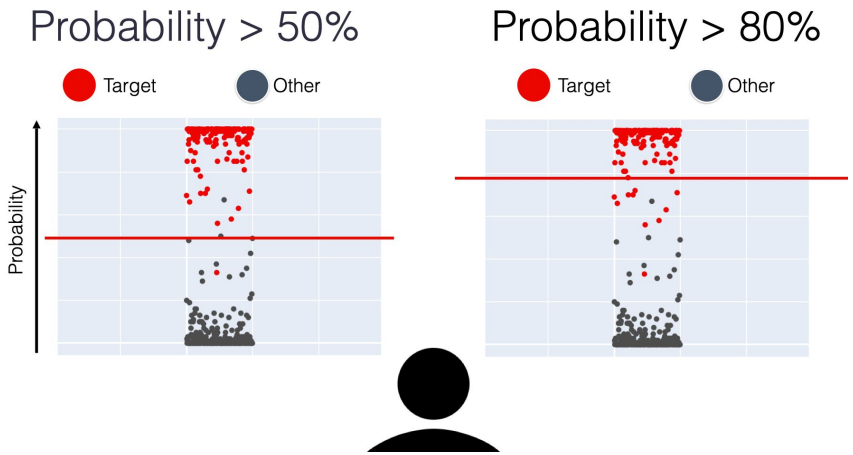

LIVE
aあ
Live Score: 33
预测: O
信心: 33%
Target: C

Data stored in JSON format = like a excel sheet in coding language

```
[server] Python result: {
[server]   ok: true,
[server]   data: {
[server]     type: 'gesture_result',
[server]     client_id: 'client_1762135533541_m4qwmoz',
[server]     hands_detected: true,
[server]     target: 'A',
[server]     predicted: 'H',
[server]     confidence: 0.3333333333333333,
[server]     landmarks_ok: false,
[server]     landmarks: [
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object],
[server]       [Object], [Object], [Object]
[server]     ],
[server]     server_ts: 1762135692159,
[server]     inference_ms: 44.9
[server]   }
[server] }
```

Field	Meaning
predicted: "H"	AI's **guess for** the gesture
target: "A"	The gesture **we wanted** it to detect
confidence: 0.33	AI's confidence level (0–1). 0.33 = "I'm not **sure**…"
landmarks_ok: false	Hand points were **not clear** → **don't trust this frame**
inference_ms: 44.9	Time **AI needed to think** (speed / performance)
landmarks: [...]	21 hand key points (AI sees **bones**, not skin)

How We Use the AI Output (Decision Logic)



To make the system stable and fair, we **don't trust every frame.**

We **check three things** before showing a result:

1) **Quality** first (Visibility ≥ 0.4: **Camera sees** the hand clearly)
If the hand keypoints are unclear → ignore this frame.

No clean signal = no decision.

2) **Confidence** threshold
If the confidence < 0.6, we don't show it.

Low confidence = "A**I isn't sure** yet."

3) **Time** stability (5-frame vote: Avoid random guesses)
The **same gesture must appear across multiple frames** (ex: 5 frames)

One lucky guess ≠ real answer.

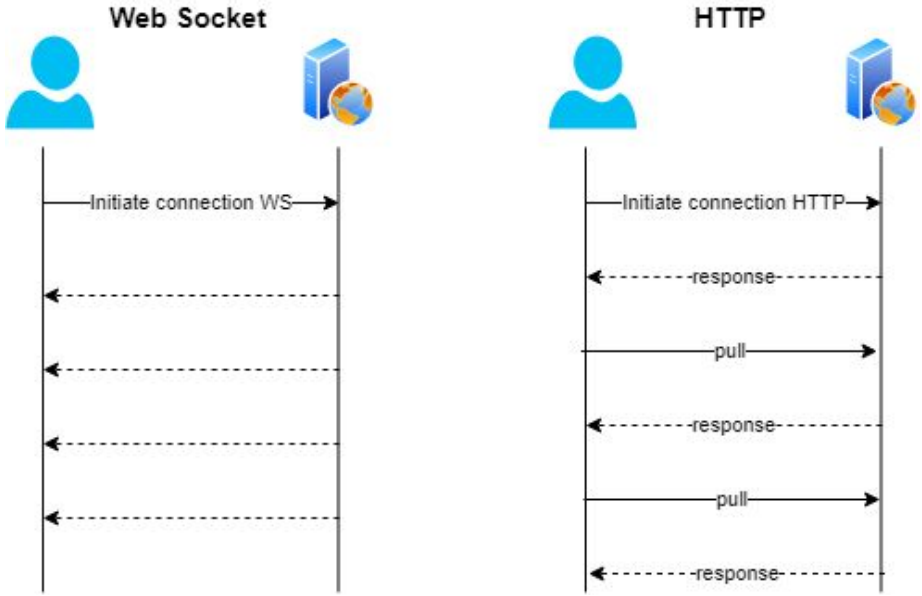
Human + AI communications: WebSocket or HTTP

HTTP (OLD telephone)

One-time request → one-time response

Connection **closes after each** message

Good for: forms, loading pages, downloading files



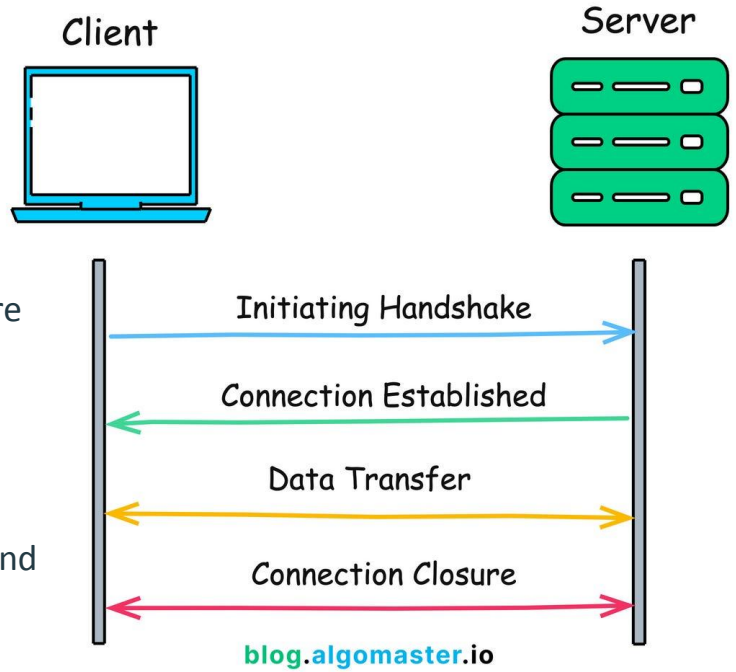
WebSockets

WebSocket(iPhone)

- Client & server connect once → **stay connected**
- Can send **messages both directions anytime**
- Good for: real-time apps (games, video chat, gesture tracking, collaboration)

Our project:

- Camera sends many hand-tracking frames per second → We need **continuous, fast communication** → **WebSocket is the better choice**



CORS Explained for Beginners

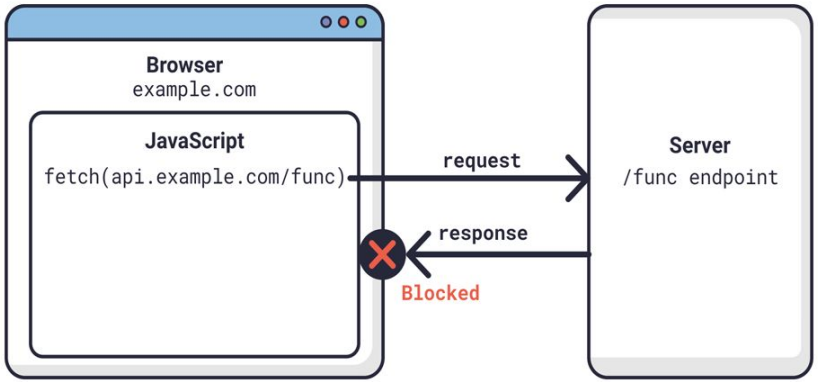


Image from haproxy.com

Cross-Origin (CORS) - Why WebSocket Sometimes Gets Blocked

Even if your front-end and back-end are both on Render, they have **different URLs**, so they count as being from "**different places**."

The browser says:

> "Are you sure you're allowed to connect?"

To fix this, your server must say:

> "Yes, I allow connections from this front-end!"

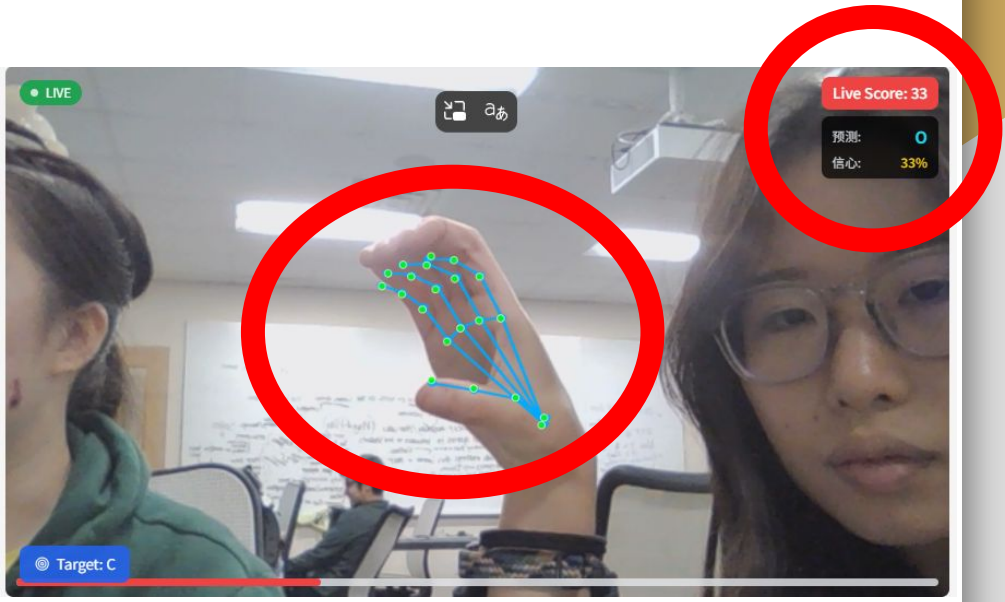
This is called CORS — a **browser safety rule.**

Try Out Our Project!

Head to the Website

https://gesture-control-xli6.onrender.com

1. At the drop-down, select ‘Webcam’
2. Click ‘Start Camera’
3. Use the ASL alphabet picture while playing with the camera.
4. Have Fun! Try A to E. 😊



Post-survey

1. Scan the QR code
2. Select ‘Teaching Sign Language with AI’ for the workshop you’re attending

We encourage you to take the survey; the data will be valuable for our research and for improving future workshops. 😄

Thanks for coming!

Any questions?